Experiment No. 3

Title

Responsive Multi-Page Website Using Media Queries

Aim

To design and develop a responsive multi-page website using media queries to ensure compatibility across different screen sizes and devices.

Software Required

- Text Editor (Notepad / VS Code)
- Web Browser (Chrome / Firefox)

Theory

Responsive web design is an approach used to create web pages that automatically adjust their layout and content based on the screen size and type of device such as desktops, tablets, and mobile phones. With the increasing use of mobile devices, it is essential to design websites that provide a consistent and user-friendly experience across all platforms.

CSS (Cascading Style Sheets) is used to control the presentation and layout of HTML documents. CSS3 introduces advanced features that help in building responsive websites, among which **media queries** play a crucial role.

Media queries allow developers to apply different CSS rules depending on device characteristics such as screen width, height, resolution, and orientation. Using media queries, the same website can adapt its layout dynamically without the need to create separate websites for different devices.

Media Query Syntax

```
@media (condition) {

    /* CSS rules */
```

}

Commonly used conditions include:

- **max-width** –This condition applies CSS styles when the screen width is less than or equal to a specified value.It is commonly used to design layouts for tablets and mobile devices.

- **min-width** – This condition applies CSS styles when the screen width is greater than or equal to a specified value.It is generally used for desktop or larger screen layouts.

In a responsive multi-page website, a single external CSS file is linked to all HTML pages. Media queries ensure that all pages maintain a consistent layout while adapting to different screen sizes, thereby improving usability and accessibility.

Procedure

1. Create a new folder and save all files inside the same folder.
2. Create multiple HTML files such as index.html, about.html, services.html, and contact.html.
3. Create a common external CSS file named style.css.
4. Link the CSS file to all HTML pages using the <link> tag.
5. Design a common navigation bar on all pages for page-to-page navigation.
6. Use CSS Flexbox for arranging content on larger screens.
7. Apply media queries in the CSS file to adjust layout for tablets and mobile devices.
8. Save all files and open index.html in a web browser.
9. Resize the browser window to observe responsive behavior.

Program Code

HTML Files

index.html – Home Page

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Home</title>
   <link rel="stylesheet" href="style.css">
```

```
</head>

<body>

<header>
   <h1>Responsive Website</h1>
   <p>Home Page</p>
</header>

<nav>
   <a href="index.html">Home</a>
   <a href="about.html">About</a>
   <a href="services.html">Services</a>
   <a href="contact.html">Contact</a>
</nav>

<div class="container">
   <div class="box">
      <h2>HTML</h2>
      <p>HTML provides structure to web pages.</p>
   </div>

   <div class="box">
      <h2>CSS</h2>
      <p>CSS is used for styling and layout.</p>
   </div>

   <div class="box">
      <h2>Media Queries</h2>
      <p>They make websites responsive.</p>
   </div>
</div>

<footer>
   <p>&copy; 2026 Web Programming Lab</p>
</footer>

</body>
</html>
```

about.html – About Page

```
<!DOCTYPE html>
<html lang="en">
```

```html
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>About</title>
   <link rel="stylesheet" href="style.css">
</head>

<body>

<header>
   <h1>About Us</h1>
</header>

<nav>
   <a href="index.html">Home</a>
   <a href="about.html">About</a>
   <a href="services.html">Services</a>
   <a href="contact.html">Contact</a>
</nav>

<div class="container">
   <div class="box">
      <h2>About Website</h2>
      <p>This website demonstrates responsive design using media queries.</p>
   </div>
</div>

<footer>
   <p>&copy; 2026 Web Programming Lab</p>
</footer>
</body>
</html>
```

services.html – Services Page

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Services</title>
   <link rel="stylesheet" href="style.css">
</head>
```

```html
<body>

<header>
   <h1>Our Services</h1>
</header>

<nav>
   <a href="index.html">Home</a>
   <a href="about.html">About</a>
   <a href="services.html">Services</a>
   <a href="contact.html">Contact</a>
</nav>

<div class="container">
   <div class="box">
      <h2>Web Design</h2>
      <p>Responsive web design services.</p>
   </div>

   <div class="box">
      <h2>Development</h2>
      <p>HTML, CSS, and basic JavaScript development.</p>
   </div>
</div>

<footer>
   <p>&copy; 2026 Web Programming Lab</p>
</footer>
</body>
</html>
```

contact.html – Contact Page

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Contact</title>
   <link rel="stylesheet" href="style.css">
</head>

<body>
```

```html
<header>
   <h1>Contact Us</h1>
</header>

<nav>
   <a href="index.html">Home</a>
   <a href="about.html">About</a>
   <a href="services.html">Services</a>
   <a href="contact.html">Contact</a>
</nav>

<div class="container">
   <div class="box">
      <h2>Get in Touch</h2>
      <p>Email: example@email.com</p>
      <p>Phone: 9876543210</p>
   </div>
</div>

<footer>
   <p>&copy; 2026 Web Programming Lab</p>
</footer>
</body>
</html>
```

CSS File

Style.css

```css
/* General Styling */
body {
   font-family: Arial, sans-serif;
   margin: 0;
   background-color: #f4f6f9;
}

header {
   background-color: #1e3a8a;
   color: white;
   text-align: center;
   padding: 20px;
}

nav {
   background-color: #2563eb;
```

```
    text-align: center;
    padding: 10px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 15px;
    font-weight: bold;
}

/* Content Layout */
.container {
    display: flex;
    justify-content: space-around;
    padding: 20px;
}

.box {
    background-color: white;
    padding: 20px;
    width: 30%;
    border-radius: 5px;
}

/* Footer */
footer {
    background-color: #1e3a8a;
    color: white;
    text-align: center;
    padding: 10px;
}

/* Media Query – Tablet */
@media (max-width: 900px) {
    .container {
        flex-direction: column;
        align-items: center;
    }

    .box {
        width: 80%;
        margin-bottom: 20px;
```

```
    }
}

/* Media Query – Mobile */
@media (max-width: 600px) {
    nav a {
        display: block;
        margin: 10px 0;
    }
}
```

Student Tasks

Task 1: Add an Active Navigation Highlight

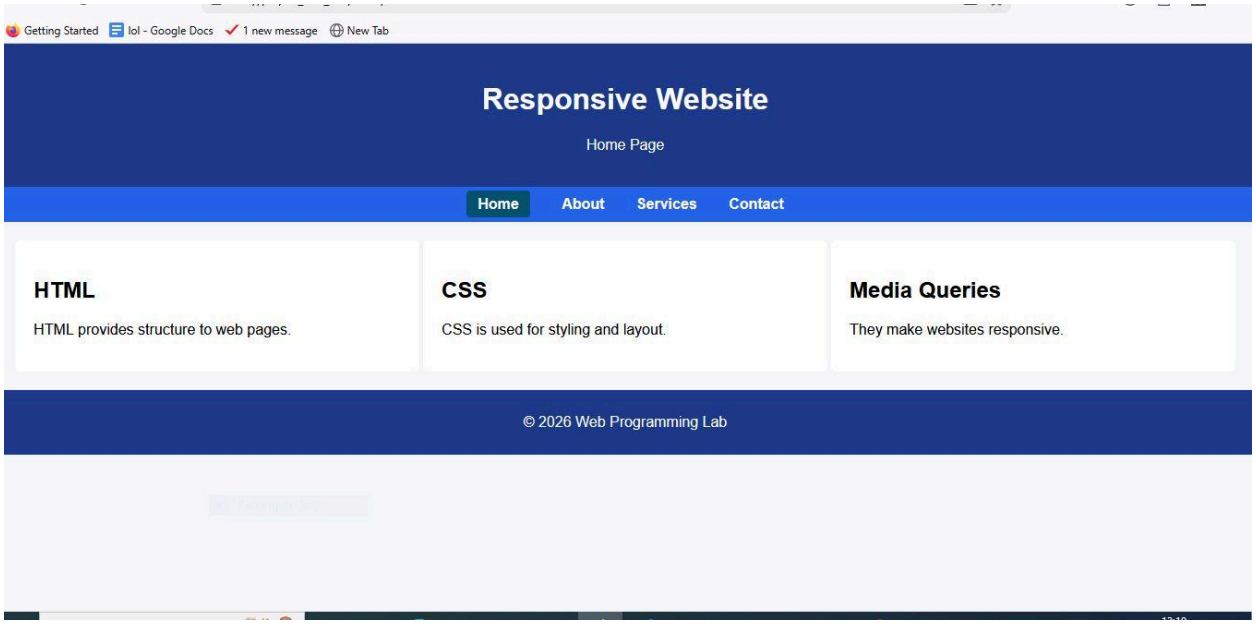
Highlight the current page in the navigation bar. Users can easily identify the current page.

Program:

```
nav a:hover {
    background-color: lightblue;
  color: rgb(8, 112, 172);
  padding:6px 12px;
  border-radius: 4px;
  transition: transform 0.2s, box-shadow 0.2s;

}
nav a.active{
    background-color: #065371;
    padding:6px 12px;
  border-radius: 4px;
}
```

Output:



Task 2: Improve Mobile Responsiveness

Add an additional media query for very small screens (≤ 480px) to reduce font size and spacing.For Better readability on small mobile devices.

Program:

```
/* Media Query – Mobile */
@media (max-width: 480px) {
   nav a {
      display: block;
      margin: 10px 0;
      font-size: small;
   }
   .img {
      width: 100%;
      height: auto;
   }
}
```

Output:

Responsive Website

Home Page

Home

About

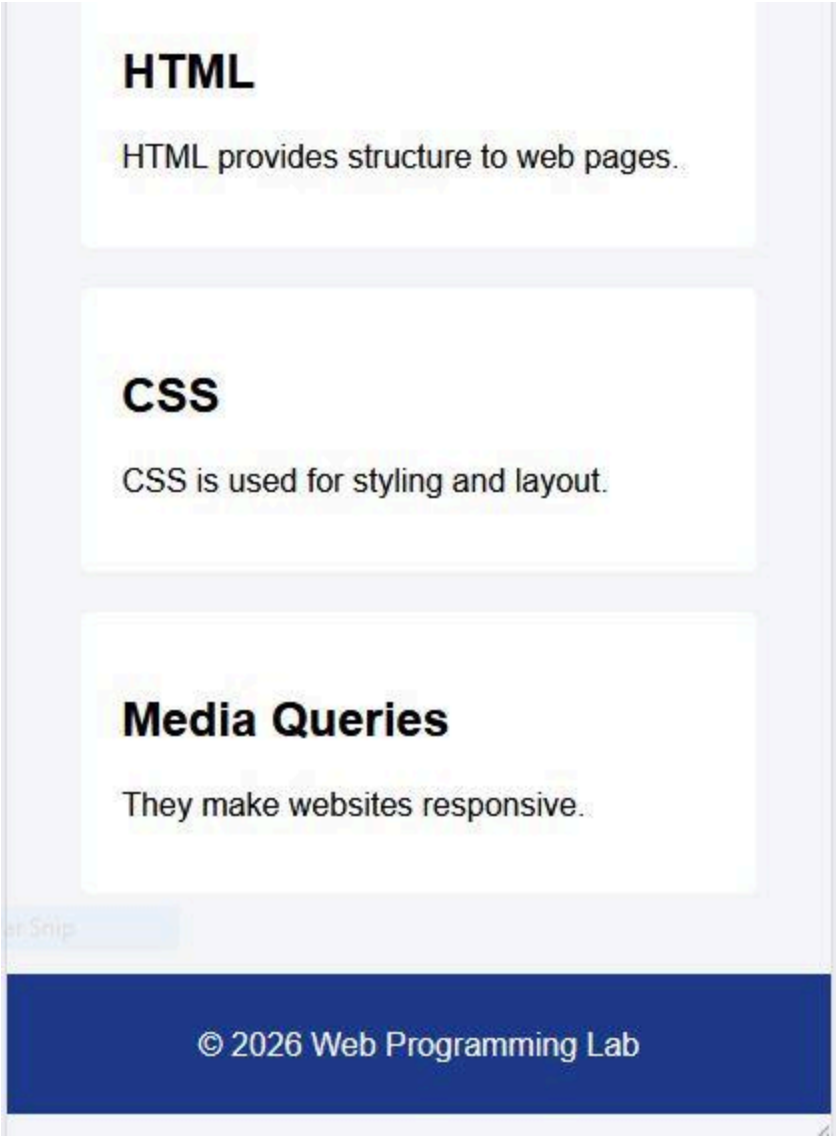
Services

Contact

HTML

HTML provides structure to web pages.

CSS



Task 3: Convert Layout to Flexbox or grid and change CSS individually for each html page.

Replace Flexbox with CSS Grid and vice versa and change designs too .

Program:

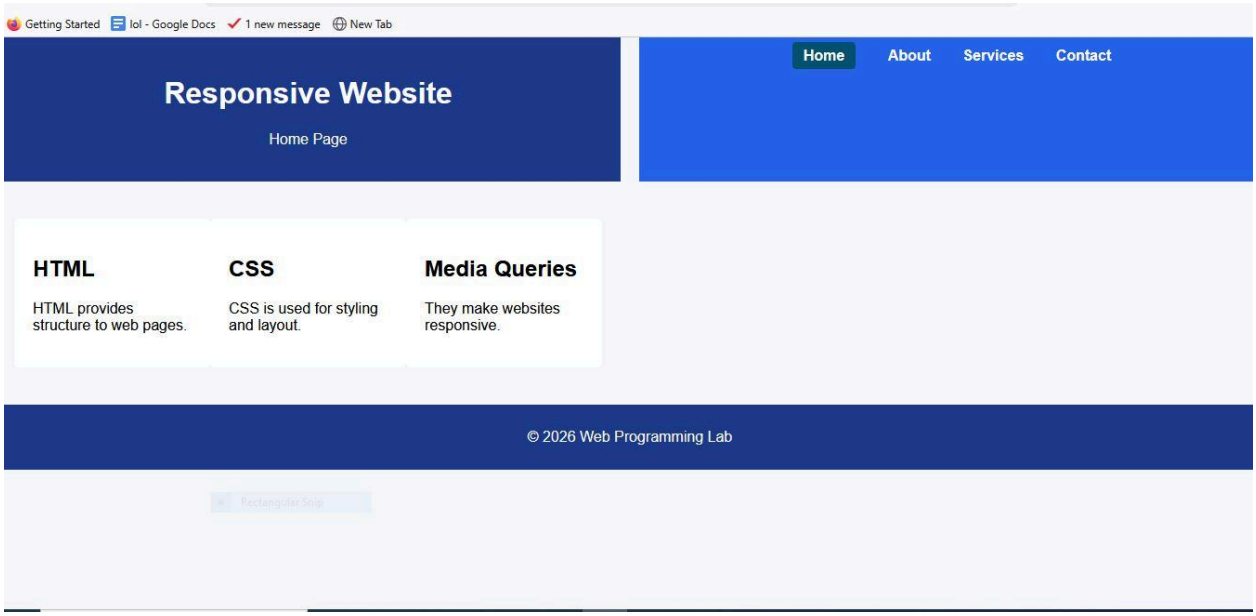
```
body {
   font-family: Arial, sans-serif;
   margin: 0;
   background-color: #f4f6f9;
     display: grid;
   grid-template-columns: repeat(2, 1fr);
   grid-template-rows: auto auto auto;
   gap: 20px;
```

```
}

header {
   background-color: #1e3a8a;
   color: white;
   text-align: center;
   padding: 20px;
}

nav {
   background-color: #2563eb;
   text-align: center;
   padding: 10px;
```

Output:



Task 4: Add a Contact Form

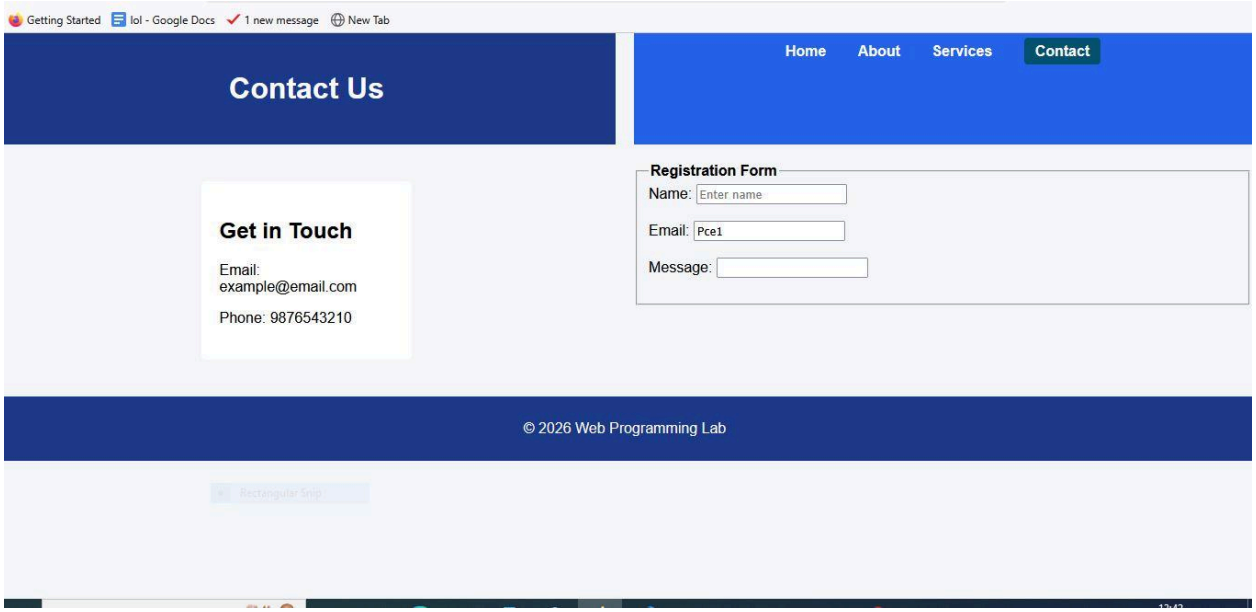
Add a contact form to contact.html with Name, Email, and Message fields.

Program:

```
<div class="container">
   <div class="box">
      <h2>Get in Touch</h2>
      <p>Email: example@email.com</p>
      <p>Phone: 9876543210</p>
   </div>
</div>
```

```
<section>
   <form>
      <form align="center">
<fieldset>
<legend><b>Registration Form</b></legend>
Name:
<input type="text" placeholder="Enter name" required>
<br><br>
Email:
<input type="email" required>
<br><br>
Message:
<input type="textbox" required>
<br><br>
   </form>
</section>
```

Output:



Task 5: Add Responsive Images

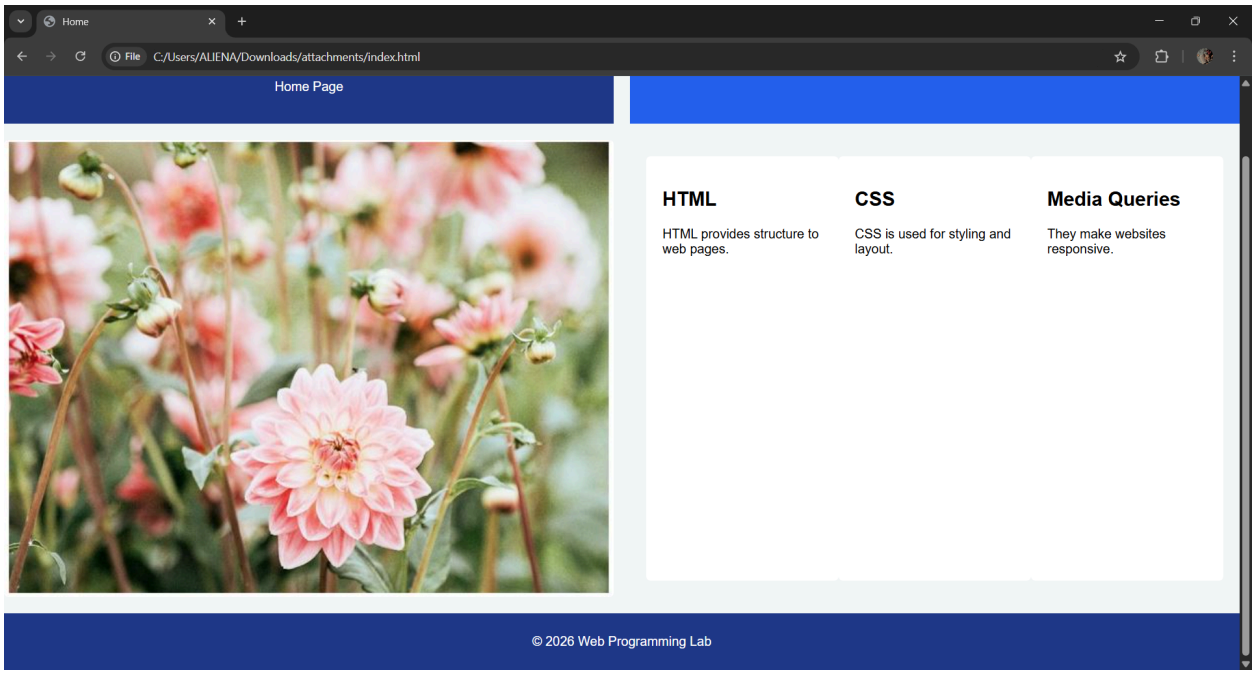
Add an image to each page and ensure it resizes on smaller screens.

Program:

```
<nav>
   <a class=active href="index.html">Home</a>
   <a   href="about.html">About</a>
   <a  href="services.html">Services</a>
```

```
    <a  href="contact.html">Contact</a>
</nav>
<img src="7.JPG" alt="Responsive Design" style="width:100%;height:auto;">
<div class="container">
    <div class="box">
        <h2>HTML</h2>
        <p>HTML provides structure to web pages.</p>
    </div>
```

Output:



Result

A responsive multi-page website was successfully designed and implemented using CSS media queries to ensure compatibility across different screen sizes and devices.